Double Integrals and Monte Carlo Integration

Integration in two dimensions with SymPy, SciPy and random sampling

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November 4, 2026

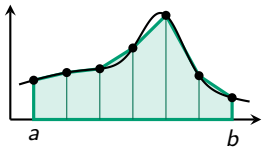
Review: integration in one dimension

Exact: SymPy

`sp.integrate(f, (x, a, b))` ✓ exact ✗ may return `Integral(...)` → `.evalf()`

Numerical: SciPy fit a simple shape on each piece and integrate it exactly

Trapezoid

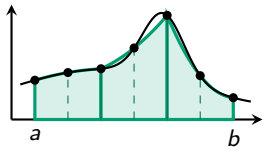


a line on each piece

`trapezoid(yv, x=xv)`

error $\sim h^2$

Simpson



a parabola on each pair

`simpson(yv, x=xv)`

error $\sim h^4$

Adaptive Gauss

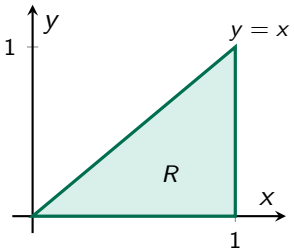


$\sum w_i f(x_i)$, split where f bends

`quad` (adaptive)

exact for degree $\leq 2n-1$

Double integrals: iterated integration



$$\iint_R xy\,dA = \int_0^1 \int_0^x xy\,dy\,dx = \frac{1}{8}$$

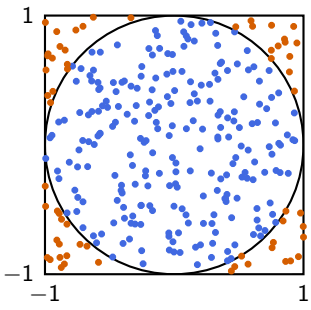
volume under $z = f(x, y)$ over R:

$$\iint_R f\,dA = \int_a^b \int_{g(x)}^{h(x)} f(x, y)\,dy\,dx$$

1. inner integral first (y, with x fixed)
2. then the outer integral (x)

Disks: polar coordinates, $dA = r\,dr\,d\theta$ (don't forget the r)

Monte Carlo integration



238 of 300 inside: $\pi \approx 4 \cdot \frac{238}{300} = 3.17$

integral = area × average of f

$$\iint_R f \, dA \;\approx\; |B| \cdot \frac{1}{N} \sum_{i=1}^{N} f(x_i, y_i)$$

1. random points in a box B around R

2. $f = 0$ outside R

3. average × box area

error $\sim 1/\sqrt{N}$, in any dimension

$|B|$ = area of the box you sampled from